1. Mulhim User Flow Documentation

✓ Purpose of this Flow

This flow represents the complete onboarding journey for a Bioinformatics user in the Mulhim application.

The objective is to transform a laboratory DNA test into personalized AI-generated health plans. Unlike a regular fitness application, Mulhim combines:

- DNA analysis
- Laboratory biomarkers
- Phenotype questionnaire
- Lifestyle questionnaire
- User goals
- to generate personalized recommendations.

2. Phase 1 — Package Purchase

The user first discovers Mulhim through social media or the landing page.

The user:

- learns about Mulhim
- selects a testing package
- purchases the package
- enters shipping information

The order is then transferred to Jadhoor to prepare the DNA collection kit.

✓ Purpose:
  Collect user information and initiate the DNA testing process.

3. Phase 2 — DNA Collection

After downloading the application:

- the collection kit is delivered
- the user provides the saliva sample

- the courier collects the kit
- the sample is sent to the laboratory
- ✓ Purpose:
  Collect biological samples required for bioinformatics analysis.

4. Phase 3 — Phenotype Collection

While laboratory processing is taking place, the application collects additional information through questionnaires.

The questionnaire collects:

- lifestyle
- health history
- nutrition habits
- physical activity
- sleep
- personal goals
- ✓ Purpose:
  The AI uses phenotype information together with DNA results to produce more personalized recommendations. Without this information, recommendations would rely only on genetics.

5. Phase 4 — AI Analysis

After laboratory analysis is completed, the system combines:

- DNA results
- Biomarkers
- Phenotype questionnaire
- Lifestyle questionnaire
- User goals
- The AI analyzes all available information to:
- identify health risks
- identify strengths

- prioritize health areas
- generate personalized recommendations
- ✓ Purpose:
  Convert biological data into personalized health insights.

6. Phase 5 — User Authentication

After the analysis is completed:

The user receives:

- SMS
- Email
- WhatsApp
- Push notification
- containing the application download link.
- The user signs in using the Bioinformatics subscription.
- ✓ Purpose:
  Allow access only after laboratory analysis has been completed.

7. Phase 6 — Personalized Onboarding

The application then completes user profiling.

- Information collected includes:
- preferred name
- body measurements
- health goals
- preferred workout location
- activity level
- nutrition assessment
- workout schedule
- plan preference
- allergies
- injuries

- medical conditions

✓ Purpose:

Fine-tune the AI recommendations before displaying the home page.

8. Phase 7 — Home

The Home page is shown only after all required information has been collected.

At this stage, the AI has enough information to generate personalized plans.

The user receives:

- Recovery Plan
- Exercise Plan
- Nutrition Plan
- Mental Health Plan
- Hair & Skin Plan
- Each plan is continuously updated according to:
- new biomarker results
- user progress
- completed habits
- updated goals
- Notes for Developers

Blue boxes represent user interaction screens.

Yellow boxes represent system processes or external services.

Red boxes indicate Bioinformatics subscription requirements.

Green sections (in the rest of the board) represent AI/Bioinformatics logic responsible for generating and updating personalized health plans.

Do not modify completed onboarding steps without checking the overall flow, as later AI decisions depend on the collected information.

9. Plans Description for User

✓ Workout Flow Explanation

This section represents the Exercise Plan within the Mulhim application.

When the user selects Workout from the bottom navigation bar, they are taken to the Workout section.

The user can:

- View today's workout.
- View the full weekly workout plan.
- Add an extra exercise manually.
- Regenerate the workout plan if their goals or health data change.
- The AI generates a personalized athletic development plan based on:
- DNA analysis
- Biomarker results
- Phenotype questionnaire
- Lifestyle information
- User goals
- The workout plan includes:
- Training focus
- Priority level
- Progress overview
- Athletic performance traits
- 12-week performance goals
- Weekly workout plan
- Daily workout
- Each workout provides:
- Exercise title
- Exercise image
- Duration
- Difficulty level
- Equipment required
- Target muscle group
- Exercise description
- Step-by-step instructions
- After completing the workout, the user marks it as done.

✓ The AI then:
    - Tracks the user's progress.

- Updates weekly goals.
- Generates new personalized workout recommendations based on the user's performance and progress.

➢ Hair & Skin Flow Explanation

When the user opens the Hair & Skin section from the bottom navigation, the AI generates a personalized beauty and hair care plan based on the user's DNA results, biomarkers, phenotype questionnaire, lifestyle, and health goals.

The system first displays AI-generated health insights, including the user's skin health score, hair health score, UV sensitivity, and hydration status.

➢ Based on this analysis, the AI creates a complete daily protocol consisting of:

- Morning skincare routine.
- Evening skincare routine.
- Hair care routine.
- Daily healthy habits (hydration, sleep, nutrition reminders).

The user is also provided with weekly goals, such as improving skin hydration, strengthening hair, reducing breakouts, and maintaining daily sunscreen use.

As the user completes their daily routines and marks tasks as done, the AI continuously tracks progress, evaluates improvements, and automatically updates future recommendations to keep the plan personalized and adaptive.

➢ Recovery & Sleep Flow Explanation

When the user opens the Recovery section, the AI generates a personalized sleep and recovery plan using the user's DNA results, biomarkers, phenotype questionnaire, lifestyle habits, and previous recovery history.

The AI first presents personalized sleep insights, including:

- Sleep Score
- Recovery Score
- Recommended bedtime
- Recommended wake-up time

- Recommended sleep duration

✓ Based on these insights, the AI creates a personalized daily sleep protocol that includes:
  - Morning routine
  - Evening routine
  - Sleep environment optimization
  - Hydration reminders
  - Screen-time reminders before bed

✓ The AI also evaluates the user's Injury Risk Profile, displaying:
- Overall risk score
- Tendon & ligament strain risk
- Fast-twitch muscle strain risk
- Joint stability baselines · Stress fracture susceptibility
- Based on this profile, the AI provides personalized prevention measures: ·
- Dynamic warm-up routines prior to training
- Eccentric loading exercises
- Targeted hydration and post-workout protein guidelines
- Lower-body recovery protocols (foam rolling, contrast showers)

After completing the daily habits, the user marks them as done.

The AI then evaluates the user's sleep quality and updates the next day's recovery plan. If poor sleep quality is detected, the AI automatically adjusts the following day's exercise intensity to reduce fatigue and improve recovery.

The system continuously tracks weekly sleep progress and generates updated AI recommendations, ensuring that the recovery plan adapts to the user's long-term progress and health status.

➢ Nutrition & Hydration Flow Explanation

When the user opens the Nutrition section from the bottom navigation, the AI generates a personalized nutrition and hydration plan based on the user's DNA results, biomarkers, phenotype questionnaire, lifestyle habits, and health goals. The AI first presents the Macro Blueprint, displaying exact daily targets:

Protein percentage

Carbohydrates percentage

Fats percentage

Daily caloric targets dynamically adjusted based on energy expenditure

Based on this analysis, the AI creates a Smart Meal Timing Strategy and 7-Day Meal Plan including: ·

- Breakfast
- Lunch
- Pre-workout meal (fast-digesting carbs 30 mins before workout)
- Post-workout meal (high-protein intake within 30 mins)
- Dinner
- Daily healthy snacks

The AI also provides a Supplement & Hydration Protocol that includes:

- Targeted supplement recommendations (Creatine, Whey Isolate, Omega-3, Vitamin D3)
- Precise dosage and timing instructions
- Daily hydration targets (e.g., 3L per day)
- Daily caffeine caps and hard cutoff times (e.g., no caffeine after 2:00 PM)

After completing daily meals and tracking hydration, the user marks them as done. The AI continuously tracks daily nutrient intake and progress, evaluating energy levels and workout performance to automatically update caloric targets and future meal recommendations.

Ø Mental Health Flow Explanation

When the user opens the Mental Health section from the bottom navigation, the AI generates a personalized mental wellness and focus plan based on the user's DNA results, biomarkers, phenotype questionnaire, stress levels, and lifestyle habits.

The AI first presents genetic stress response insights, including:

- Overall genetic stress sensitivity score
- Peak cognitive focus windows (e.g., 2:00 PM - 5:00 PM)
- Neuro-genomic fatigue thresholds

Based on these insights, the AI creates a daily mental practice protocol that includes:

- Morning meditation sessions to lower baseline cortisol
- Interactive Pomodoro focus tools (25-minute work intervals)
- Guided Box Breathing exercises (4-4-4-4 technique) for stress mitigation

- Nature exposure tracking
- Evening journaling routines to reduce cognitive load
- Screen-free digital detox windows prior to sleep

The AI also provides a Targeted Mental Support Supplement Protocol including:

- Omega-3 (DHA/EPA) for cognitive function
- Ashwagandha for stress resilience
- L-Theanine for calm focus
- Precise dosage and timing recommendations for each supplement

After completing the daily mental exercises and habits, the user marks them as done.

The AI continuously tracks daily mental wellness progress, cross-referencing stress logs with sleep and workout recovery data. If high stress or CNS fatigue is detected, the AI automatically triggers immediate relaxation protocols and updates future recommendations to maintain calm focus and cognitive performance.

10. Coach & Bioinformatics Dashboard Flow Explanation

When the coach logs into the platform, they are taken to the Coach Dashboard where they can manage clients, review AI-generated recommendations, and access detailed bioinformatics reports.

The coach can select between two main pathways:

- Athlete Path (for general fitness coaches)
- Bioinformatics Path (for specialized data and research analysis)

Under the Athlete Path, the coach manages client plans through the following steps:

- Access the Clients List.
- Filter clients based on whether they have a Juthoor Health Report or an active plan.
- Display Client Profile and retrieve Phenotype Data collected from Mulhim questionnaires.
- Review AI Recommendations (generated based on the health report).
- System alerts or email notifications prompt the coach when a new AI report is ready for review.
- Modify recommendations and add custom Coach Notes if needed.
- Grant Coach Approval to finalize the plan.

Once approved, the system generates the Final Personalized Plan, and the coach chooses the Export & Branding Option:

- Add Logo Option: Applies the Mulhim Logo to the generated plan (Standard / Free).
- Do Not Add Logo Option: Removes the Mulhim Logo for a white-label plan (Charges an Extra Fee to the coach's account).

After exporting, the plan is sent directly to the client:

- Send Plan to Client.
- Send Approved Plan Notification to the user.
- Provide a communication channel for direct feedback between coach and user on performance or AI/Bioinformatics utilization.

Under the Bioinformatics Path, the specialist accesses technical report analysis:

- Access Interactive Learning resources.
- View the complete Health Report provided by Juthoor.
- Mulhim AI determines which specific technical parts of the report are displayed to the coach.
- Display Report Insights derived from research papers and Juthoor data for in-depth genomic analysis.

11. Mulhim Categories:

- Recovery (Sleep recovery)
- Nutrition
- Exercise
- Hair and Skin
- Mental Health

12. Juthoor Categories:

➢ Recovery
  ✓ Features Available in Juthoor: Daily Recovery Actions
- The platform provides:
  - Lower-Body Recovery Protocol: Step-by-step guidance including foam rolling, contrast showers, and prioritizing full sleep duration.
  - Active Recovery: Low-intensity movements and light stretching to increase blood flow, alleviate muscle soreness, and speed up tissue repair.
  - Injury Risk Profile
- Juthoor evaluates injury susceptibility by displaying:
  - Overall risk level (e.g., 45% moderate risk score)
  - Tendon & ligament risk (collagen elasticity and strain risk)
  - Muscle strain risk (fast-twitch vulnerability)
  - Joint stability (range of motion and stability baselines)
  - Stress fractures (bone mineral density and fracture susceptibility)
  - Personalized Prevention Measures
- The platform includes:
  - 10-12 minute dynamic warm-up routines prior to explosive movements.
  - Eccentric loading exercises twice weekly for hamstrings and Achilles tendons.
  - Hydration goals (e.g., 3L/day) paired with 30g of protein within 30 minutes post-workout.
- Strengths
  - Comprehensive daily recovery guidance.
  - Detailed biomechanical risk profiling.
  - Structured preventive measures.
  - Practical actionable protocols.

➢ Opportunities for Improvement in Mulhim
  ✓ Mulhim will improve the Recovery experience by:
  - Generating recovery plans based on genetic results, biomarkers, phenotype data, and user goals.
  - Adjusting recovery recommendations dynamically according to daily user progress and strain.
  - Integrating computer vision for real-time posture and mobility scanning during recovery routines.
  - Using the Mulhim AI assistant to explain injury risks and recovery steps in simple language.

➢ Personalized Nutrition & Hydration Plan

- ✓ Features Available in Juthoor: Macro Blueprint (Optimal Macronutrient Distribution)
- The platform calculates exact macro ratios based on genetic profile and fitness goals:
    - Protein: 35%
    - Carbohydrates: 45%
    - Fats: 20%
    - Smart Meal Timing Strategy

- The platform outlines:
    - Pre-Workout (30 min before): Fast-digesting carbs and light hydration (e.g., banana or espresso).
    - Post-Workout (within 30 min): High-protein intake for muscle glycogen replenishment and repair.
    - Evening Meal: Balanced meal supporting overnight muscle recovery.
    - 7-Day Meal Plan
- The platform generates:
    - Dynamic caloric adjustments based on daily output (e.g., 2,850 kcal on High-Intensity Training Days).
    - Precise daily schedules for Breakfast, Lunch, Pre-Workout, Dinner, and Snacks.
    - Supplement & Hydration Protocol
- The platform provides:
    - Targeted Supplements: Dosage recommendations for Creatine Monoglyceride, Whey Protein Isolate, Omega-3 Fish Oil, Vitamin D3.
    - Caffeine & Hydration Caps: Daily caffeine cap (e.g., 400 mg) with a hard cutoff time (e.g., no caffeine after 2:00 PM).
- Strengths
    - Precise macronutrient breakdowns.
    - Structured meal timing and 7-day plans.
    - Comprehensive supplement and hydration guidelines.
    - ✓ Opportunities for Improvement in Mulhim
- Mulhim will improve the Nutrition & Hydration experience by:
    - Customizing nutrition plans using a multi-omics approach combining genetics, live biomarkers, and lifestyle data.
    - Adapting meal plans and caloric targets automatically based on real-time activity and progress tracking.
    - Providing smart, contextual hydration and nutrition reminders through the AI assistant.
- ➢ Mental Wellness & Performance
    - ✓ Features Available in Juthoor: Genetic Stress Response
- The platform evaluates:
    - Genetic sensitivity to stress (e.g., displaying a 58% moderate sensitivity score).
    - Focus & De-stress Tools
- The platform recommends:

- Peak Focus Windows: Identifies daily peak focus hours (e.g., 2:00 PM - 5:00 PM).
- Pomodoro Focus: Structured 25-minute work intervals followed by 5-minute breaks.
- Box Breathing: 4-4-4-4 breathing technique to regulate the autonomic nervous system.

- Daily Mental Practice Protocol
  - Morning Meditation: 10-minute session post-waking to lower baseline cortisol.
  - Journaling: 5-minute evening routine before bed to clear cognitive load.
  - Nature Exposure: 20 minutes daily to boost mood and circadian alignment.
  - Evening Digital Detox: 60-minute screen-free window prior to sleep.
  - Mental Support Supplements: Doses of Omega-3 (DHA/EPA), Ashwagandha, and L-Theanine.
- Strengths
  - Clear focus and productivity tools.
  - Structured daily mental practice protocols.
  - Targeted mental support supplementation.

➢ Opportunities for Improvement in Mulhim

- Mulhim will improve the Mental Wellness experience by:
  - Combining genetic stress profiles with live wearable biomarkers and sleep quality data.
  - Adapting mental wellness and focus recommendations based on real-time fatigue and stress scores.
  - Delivering interactive guidance and reminders through the Mulhim AI health coach.

➢ Exercise & Sports

✓ Features Available in Juthoor:

- Athletic Development Plan: The platform generates a personalized training plan based on the user's genetic report.

- It includes:

- Training focus (Power & Strength)
- Genetic summary
- Priority level
- Training progress bar
- Injury Risk Profile
- Jadhoor evaluates injury susceptibility by displaying:
- Overall risk level
- Genetic risk percentage
- Tendon & ligament risk
- Muscle strain risk
- Joint stability

- Stress fracture risk
- It also provides personalized prevention recommendations.
- Genetic Fitness Traits
- The platform evaluates multiple athletic characteristics, including:
- Explosive power
- Strength potential
- Endurance
- Recovery rate
- Flexibility
- VO_2 Max
- Agility
- Balance
- Pain tolerance
- Bone density
- Sprint potential
- Muscle composition
- Each characteristic is displayed using progress indicators.
- Performance Goals
- Juthoor creates measurable goals over a fixed period.

* Current duration:
  - 12-week performance goals
  - Weekly Training Plan
  - The platform generates a complete weekly schedule.
  - Each day contains:
  - Exercise title
  - Exercise image
  - Training duration
  - Difficulty level
  - Equipment required
  - Muscle group
  - Exercise description
  - Step-by-step instructions
  - "Done" button
  - Users can switch between:
  - Today's workout
  - Full weekly plan
* Strengths
  - Personalized weekly exercise program.

- Clear daily schedule.
- Goal-based planning.
- Exercise completion tracking.
- Simple visual presentation.
- ➢ Opportunities for Improvement in Mulhim
  - ✓ Mulhim will improve this experience by:
- Generating plans based on genetic results, biomarkers, phenotype data, and user goals, not genetics alone.
- Adjusting plans dynamically according to user progress.
- Allowing AI to regenerate plans whenever goals change.
- Using the Mulhim mascot as an interactive AI health coach.
- Providing motivational messages and adaptive recommendations throughout the health journey.

5. Sleep

- ➢ Features Available in Juthoor
- Personalized Sleep Window
- The platform recommends:
- Bedtime
- Wake-up time
- Sleep duration
- Chronotype
- Melatonin efficiency
- It also displays a visual sleep timeline.
- Nightly Habit Protocol
- The platform provides personalized habits including:
- Sleep schedule
- Evening wind-down routine
- Morning routine
- Sleep environment
- Melatonin supplementation
- Magnesium
- L-Theanine
- Screen restriction before bedtime
- Each recommendation includes:
- Dose
- Timing

- Application instructions
- Completion button

- Strengths
  - Personalized bedtime schedule.
  - Daily sleep protocol.
  - Easy-to-follow recommendations.
  - Habit tracking.
    - ✓ Opportunities for Improvement in Mulhim
  - Mulhim will enhance the experience by:
  - Updating recommendations based on the user's daily sleep quality.
  - Modifying exercise and recovery plans automatically after poor sleep.
  - Delivering personalized reminders through the Mulhim AI assistant.
  - Explaining recommendations using simple health insights instead of technical genetic terminology.

6. Hair & Skin (Radiant Glow)

- ➢ Features Available in Juthoor: Skin Analysis
  - ✓ The platform evaluates:
  - Collagen
  - Antioxidants
  - UV sensitivity
  - Results are displayed using progress bars.
  - Daily Beauty Protocol
  - The platform generates a skincare routine including:
  - Gentle cleanser
  - Vitamin C serum
  - Moisturizer
  - Sunscreen
  - Double cleansing
  - Retinol
  - Hyaluronic acid
  - Shampoo
  - Scalp treatment
  - Heat protection
  - Each recommendation includes:
  - Amount
  - Timing

- Application instructions
- Supplements
- Recommended supplements include:
- Biotin
- Collagen peptides
- Vitamin E
- Zinc
- Each supplement specifies:
- Dose
- Timing
- Long-Term Goals
- The platform creates measurable goals over:
- 60 days
- Examples include:
- Skin hydration
- Hair strength
- Breakout reduction
- Daily sunscreen adherence

- Strengths
  - Comprehensive daily skincare routine.
  - Personalized supplement recommendations.
  - Long-term measurable goals.
  - Structured daily guidance.

✓ Opportunities for Improvement in Mulhim

- Mulhim will improve the Hair & Skin experience by:
  - Combining genetic analysis with phenotype information and user lifestyle.
  - Providing adaptive recommendations according to user progress.
  - Delivering contextual reminders, such as sunscreen reminders on high UV days.
  - Using the Mulhim AI assistant to explain recommendations in simple language.
  - Making the experience more engaging through interactive progress tracking and motivational feedback.

13. Overall Benchmark Summary

Category	Feature	Juthoor	Mulhim
Exercise & Sports	Personalized Plans	Generated based on static genetic report only.	Multi-omics approace Genetics + live biomarkers + phenotype + goals.
Exercise & Sports	Weekly Exercise Plans	Fixed 12-week static structure.	Dynamic AI-adjusted workout plans (updates on goal/progress change).
Recovery & Sleep	Sleep & Recovery Protocol	Static bedtime schedule and pre-set habits.	Adaptive recommendations based on daily sleep quality and wearable HRV.
Hair & Skin	Hair & Skin Routine	Static skincare and supplement protocols.	Personalized and interactive guidance (e.g., UV alerts, lifestyle tracking).
All Categories	Long-Term Goals	Fixed timeframes (e.g., 12 weeks / 60 days).	Flexible, AI-generated goals that evolve with continuous progress.
All Categories	Progress Tracking	Manual check-ins and completion buttons.	AI-driven progress monitoring with Computer Vision & real-time analytics.
All Categories	Daily Recommendations	Pre-scripted advice and rigid schedules.	Adaptive and personalized daily recommendations.
AI Engagement	AI Interaction	Limited to basic interface feedback.	Full AI health companion featuring the interactive Mulhim mascot.

Content Delivery	Educational Approach	High technical genetic terminology.	Minimal jargon; replaced with practical, simple health insights.

Conclusion

The Juthoor platform provides well-structured personalized protocols for exercise, sleep, and hair & skin care based on genetic reports. However, its interface focuses heavily on presenting genetic information and fixed protocols.

Mulhim aims to build upon these strengths by creating a more user-centered experience that combines genetic results, biomarkers, phenotype data, lifestyle, and personal goals. Instead of emphasizing scientific genetic terminology, Mulhim will deliver simple, actionable health insights, adaptive AI-generated plans, and continuous guidance through the Mulhim AI assistant, creating a more engaging and practical digital health experience.